The Dataverse Project & and the Global Dataverse Community Consortium

A FAIR, Open Source Platform and Community for Sharing and Preserving Research Data

10 February 2026



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About the Presenters



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UiT The Arctic University of Norway

Agenda

- Introduction to the Dataverse Project
- Research Data Sharing & the Repository Ecosystem
- Dataverse in Practice
- The Global Dataverse Community Consortium (GDCC)
- Getting Involved

The Dataverse Project

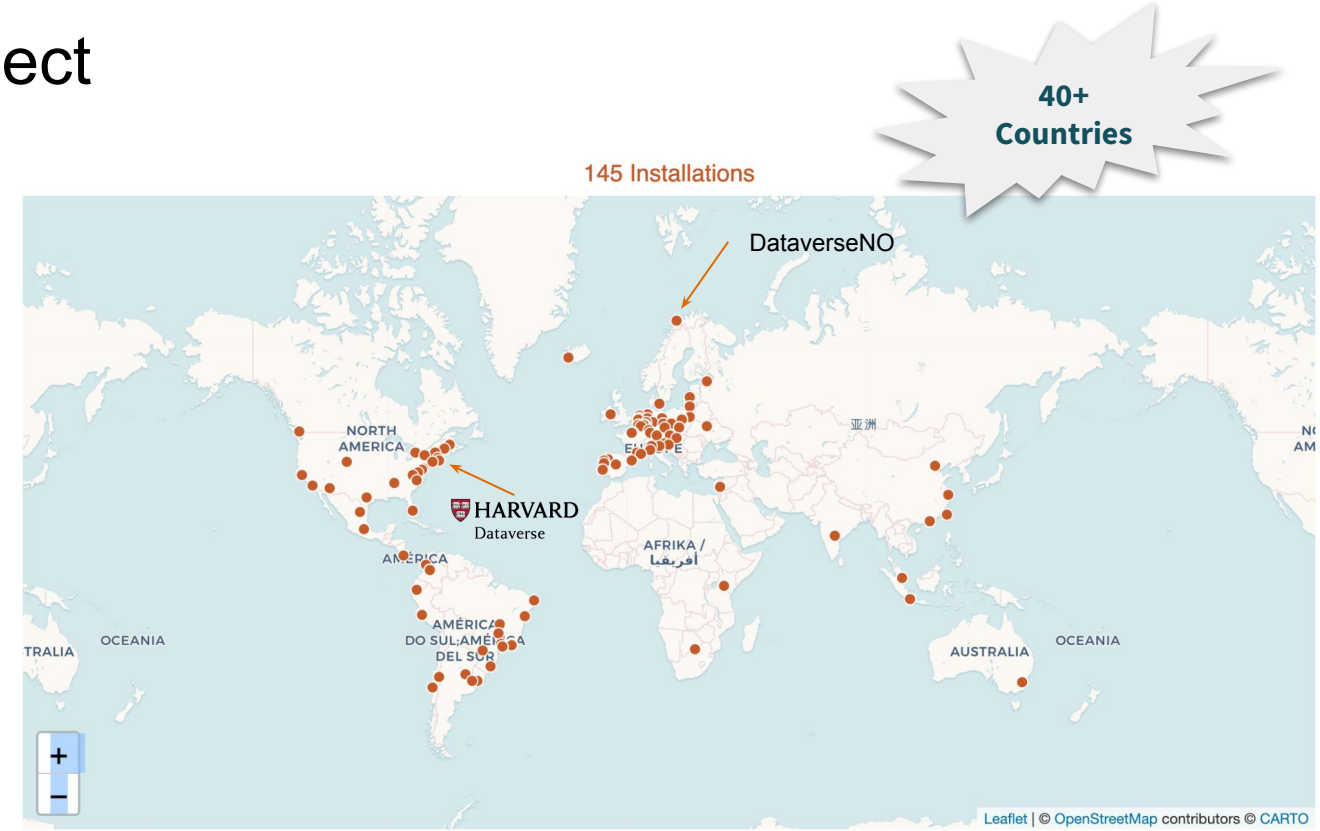
Background, Goals & Timeline

- Open Source generalist repository platform
- 141 Installations worldwide
- Founded in 1997
- Supports FAIR Data Principles

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Dataverse Project

- Open Source
- **Data Repository Software Platform**
- 145 Installations
- Active Community
- FAIR support
- Standards support
- Many features
- Previewers
- External tools
- Integrations



Dataverse Project Timeline

Dr. Gary King
Founder & Principal Investigator
Albert J. Weatherhead III University Professor
Harvard University
Institute for Quantitative Social Sciences



2025 Community Meeting, UNC

1997 ➡ 2006 ➡ 2007 ➡ 2008 ➡ 2015 ➡ 2018 ➡ 2023 ➡ 2024 ➡ 2025 ➡ 2026

1997
Project Begins

Focus on preserving & sharing Social Sciences data

2006
The Dataverse Network 1.0

🎉 Harvard Dataverse Repository established

2007
🎉 Announcing the Dataverse Network

2008
Odum Archive: First Dataverse outside Harvard

2015
Data Citations

🤝 First Open Dataverse Community Meeting

Dataverse 4.0

2018
📣 GDCC established

2023
🎉 100th Dataverse installation

Dataverse 6.0

Dataverse Re-architecture

2024
🤖 Ask the Data

🥐 Croissant

New Dataset Types (e.g., software, computational workflows)

2025
137 Dataverse Installations

Dataverse 6.7

Large Data Services

Traditional Knowledge (TK) Labels

Dataverse MCP Server

Single-Page App Launch

2026
🚀 The Next Frontier

Locally FAIR Data

Trusted Data Labels

📣 145 Dataverse Installations

🌍 Geospatial improvements

Dataverse is an easy & FAIR way to share and get credit for your research data

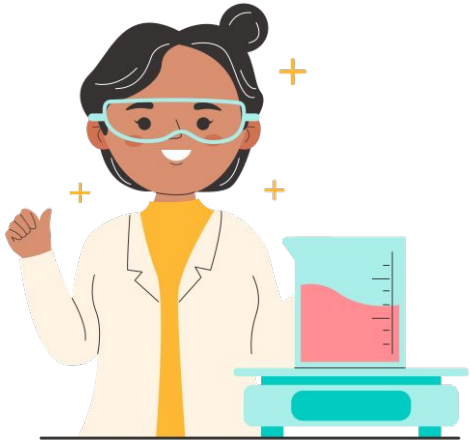


Image: Flaticon

FAIR Guiding Principles

“All research objects should be Findable, Accessible, Interoperable and Reusable (FAIR) both for machines and for people.”

Wilkinson, et al. (2016). The FAIR guiding principles for scientific data management and stewardship.

Research Data

Data Sharing & Repository Ecosystem

- What is research data?
- What is a research data repository?
- How research data repositories support FAIR
- Benefits of sharing in a repository

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Research data

“Data refers to entities used as evidence of phenomena for the purposes of research or scholarship.”

“Data are not pure or natural objects with an essence of their own. They exist in a context, taking on meaning from that context and the perspective of the beholder.”



—Borgman, C. (2015). Big data, little data, no data: scholarship in the networked world. Cambridge, Massachusetts: MIT Press.

Research data across the disciplines

	Sciences & Engineering	Social Sciences	Humanities
Evidence	Observations Measurements Samples	Observations Measurements	Primary sources Trace information
Source	Instruments Field observations	Surveys, Interviews Field observations	Art, artefacts Documents, manuscripts
Format	Binary content Images Lab notebooks Spreadsheets Samples, specimens Tabular files Streaming content	Audio recordings Diaries Images Spreadsheets Transcripts	Audio recordings Books, paper Images Maps, GIS Text files 3D

Data sharing

Safeguarding your research data and making it **accessible** to members of the research community for **use and reuse**.

- Publish, share and preserve your research data in a data repository (like Harvard Dataverse)
- **Data repositories** support dataset citations and versioning
- Support **reproducible science** by sharing data, code, and clear documentation



Image: Ainsley Seago

Data sharing benefits

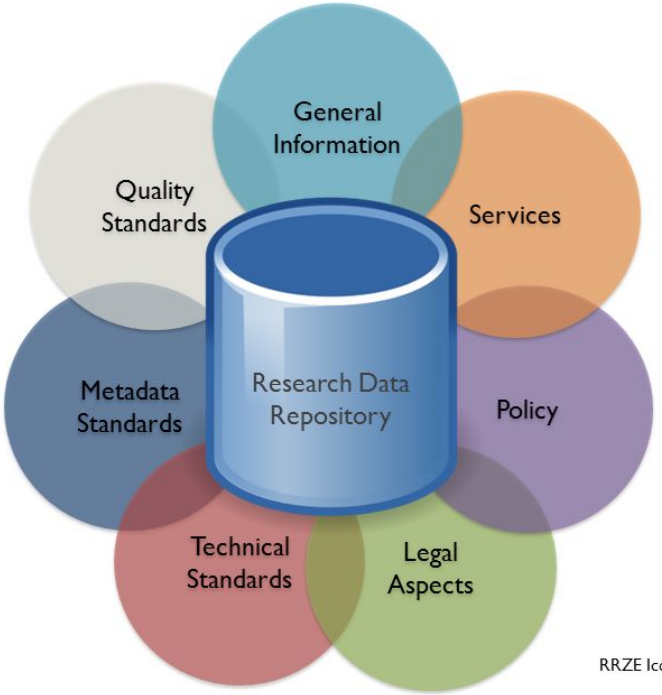
- Get credit (citations)
- Funder requirements
- Gain new insights from existing data (data reuse)
- Support research efficiency & reproducibility
- Preserve & safeguard data
- Serve long-tail of science



Image: Gerd Altmann

Data sharing in research data repositories

- Research data repositories are databases of **well-described and well-documented** research data datasets.
- Research data repositories focus on providing the **sustainable infrastructure** for the long term storage and access to, specifically, research data.
- **Research data sharing** is the practice of safeguarding research data and making it accessible to members of the research community for use and reuse. RDRs support these goals.



RRZE Icon Set (CC: BY-SA)

FAIR Guiding Principles and research data repositories

"All research objects should be Findable, Accessible, Interoperable and Reusable (FAIR) both for machines and for people." (Wilkinson, et al., 2016)

Findable	Accessible	Interoperable	Reusable
• Well-described using standards • Discoverable via aggregators & catalogs	• Persistent identifier • Citable • Reliable storage	• Standard & open formats, where possible • Formats observe good preservation practices	• Clear terms of use • Accompanied by supplementary materials • Version control

Metadata	Technical Infrastructure	Context, Policy & Rights

Research data repositories

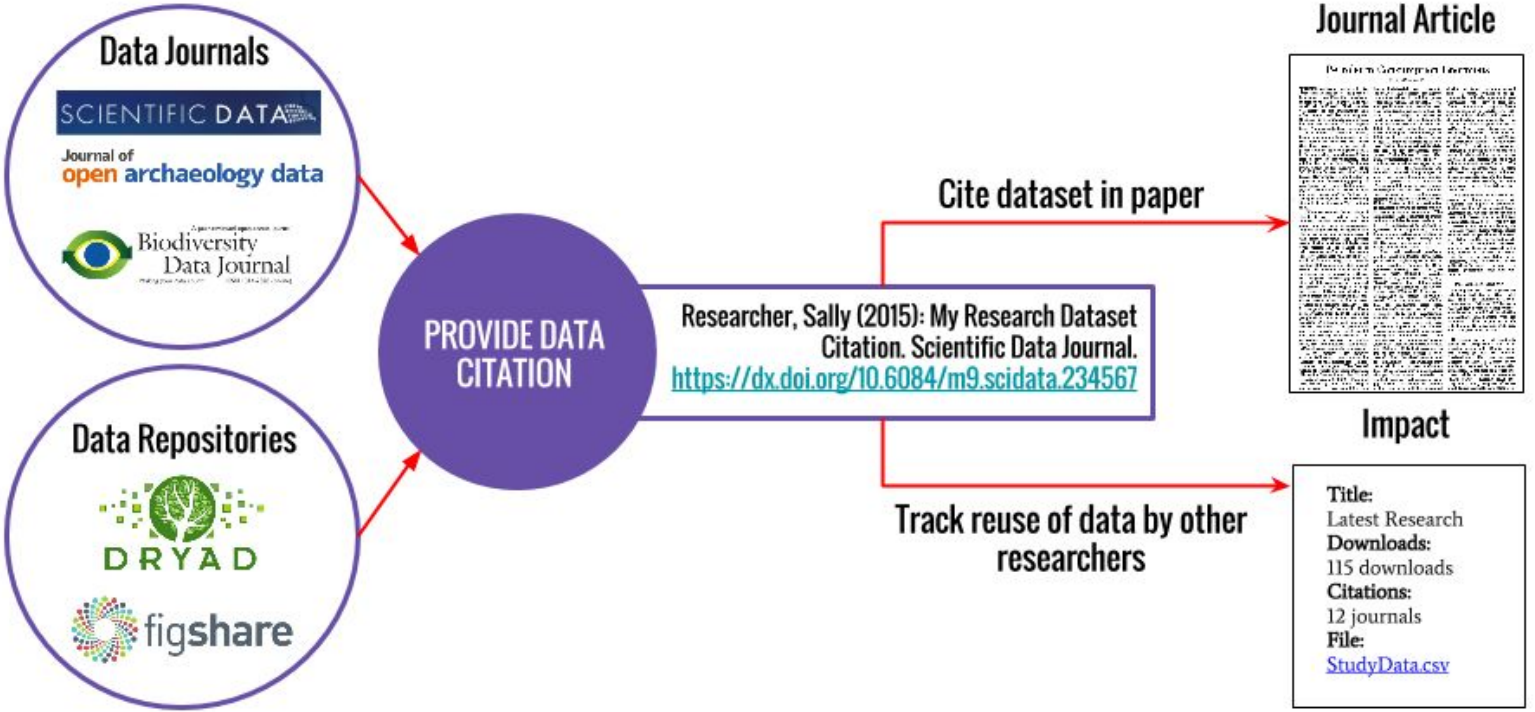
re3data.org
REGISTRY OF RESEARCH DATA REPOSITORIES
Lists 3460+ repositories!

Repository Types	• Funder • Disciplinary • Institutional • Generalist • National
Costs	• Vary • Free - paid services
Requirements & Restrictions	• Some are open • Some require affiliation • Few allow sensitive data

ROPER CENTER For Public Opinion Research
ICPSR Sharing data to advance science
OSF
zenodo
tDAR The Digital Archaeological Record
A SERVICE OF DIGITAL ANTIQUITY
DRYAD
Vivli CENTER FOR GLOBAL CLINICAL RESEARCH DATA
GEO Gene Expression Omnibus
The Dataverse® Project
figshare
Mendeley Data

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Connecting research objects in the scholarly ecosystem



Slide courtesy of Vick Steeves. “Building Services Around Reproducibility & Open Scholarship.” https://osf.io/pv6ea/

Dataverse in Practice

- Key features of the Dataverse software platform
- How data users, analysts, and researchers can find, cite, and reuse data
- Focus on **Harvard Dataverse**

Dataverse Repository Platform Features

- Support for FAIR Data Principles
- Data citation for datasets and files
- OAI-PMH (Harvesting)
- APIs for interoperability and custom integrations
- API client libraries
- DataCite integration
- Login via Shibboleth
- Login via ORCID, Google, GitHub, or Microsoft
- Login via OpenID Connect (OIDC)
- Internationalization
- Versioning
- Restricted files
- Embargo
- Custom licenses
- Custom terms of use
- Publishing workflow support

- File hierarchy
- File previews
- Preview and analysis of tabular files
- Usage statistics and metrics
- Guestbook
- Fixity checks for files
- File download in R and TSV format
- Faceted search
- Customization of collections
- Private URL
- Widgets
- Notifications
- Schema.org JSON-LD
- External tools
- External vocabulary
- Dropbox integration
- GitHub integration

- Integration with Jupyter notebooks
- User management
- Curation status labels
- Branding
- Backend storage on S3 or Swift
- Direct upload and download for S3
- Export data in BagIt format
- Post-publication automation (workflows)
- Pull header metadata from Astronomy (FITS) files
- Provenance
- Support for rsync
- Auxiliary files for data files

https://dataverse.org/software-features

Dataverse Repository Platform Integrations & APIs

Integrations

- Getting Data In (e.g., GitHub, Dropbox, RSpace, RedCap, iRODS, Open OnDemand)
- Embedding Data on Websites (DV widget)
- Analysis and Computation (e.g., Data Explorer, Binder, JupyterHub, Renku, Rclone)
- Discoverability (e.g., Geodisy, DataONE)
- Research Data Preservation (e.g., Archivematica, BagIt)

https://guides.dataverse.org/en/latest/admin/integrations.html

APIs

Installations choose which APIs to make publicly accessible

- Native API
- Search API
- Data Access API (SWORD)
- Metrics APIs

https://guides.dataverse.org/en/latest/api/index.html

pyDataverse

- Python module for Dataverse.
- It helps to access the Dataverse API's and manipulate, validate, import and export all Dataverse data-types (Dataverse, Dataset, Datafile).
- Object-oriented, dynamically generates Python objects compatible with Dataverse metadata objects.
- Supports export and import of datasets to/from different data formats

https://github.com/gdcc/pyDataverse

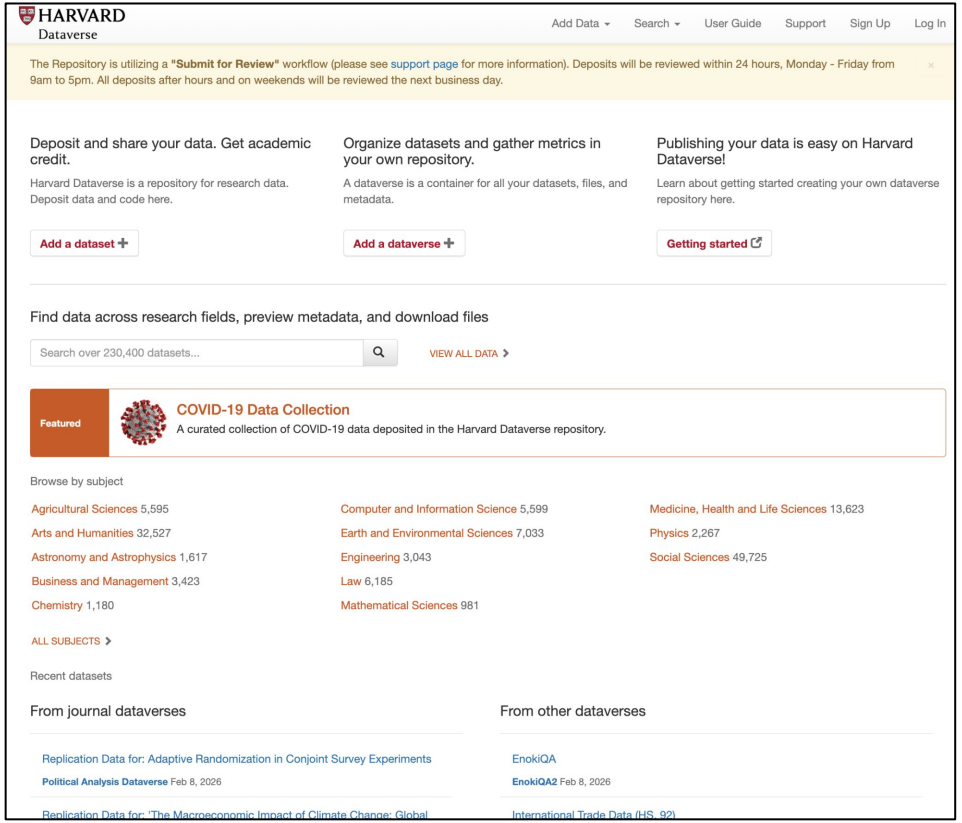
Dataverse AI-Related Resources, Tools & Projects

Resource, Tool or Project	Type	Improves	Status
AI Guide	Resource	User experience	Production
Ask the Data	Tool	Data reusability	Production
Ask Dataverse	Tool	User experience	Experimental
AutoSage	Tool	Metadata quality, User experience	Experimental
Croissant 🥐	Tool	Data findability	Production
Enhancing Dataset Metadata Project	Project (Tool)	Metadata quality, User experience	Development
GREI AI Taxonomy	Resource	Understanding of AI roles in RDM	Production
Research Data Metadata Knowledge Graph	Project (Tool)	Data findability, Data reusability	Development
Model Context Protocol (MCP) Server	Tool	Data findability, Data reusability	Production
Spam Detection & Workflow Automation	Project (Tool)	Repository data quality	Investigation
TurboCurator	Tool	Metadata quality, User experience	Production

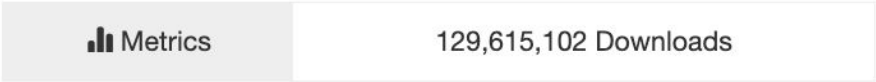
Harvard Dataverse Repository

HARVARD Dataverse

- Free endowed data repository open to all researchers, made available in 2006
- Collaboration between Harvard Library, Harvard University Information Technology, and IQSS
- All file types and disciplines are welcome
- We do not currently support sensitive data. All data must be de-identified.



https://support.dataverse.harvard.edu/



HARVARD Dataverse

Harvard Dataverse Repository

- Published Collections: 8, 492
- Published datasets: 230,380
- Published files: 5,177,187

Dataverse Category

- Research Project (3,177)
- Researcher (2,672)
- Organization or Institution (646)
- Research Group (623)
- Laboratory (185)
- Journal (154)
- Department (92)
- Teaching Course (69)

Metadata Source

- Harvested (119,669)
- Harvard Dataverse (119,203)

Harvard Dataverse storage

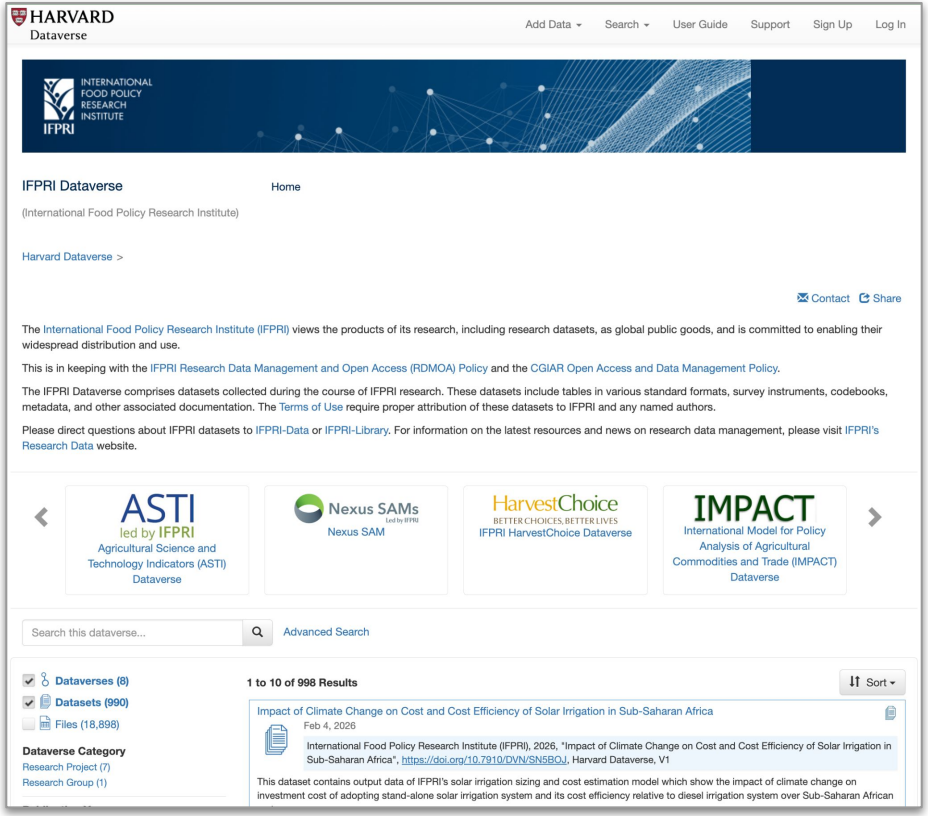
Harvard Users	Other Users	Large Data
2.5T per account free	1T per account free	Fee-based tape & disk options

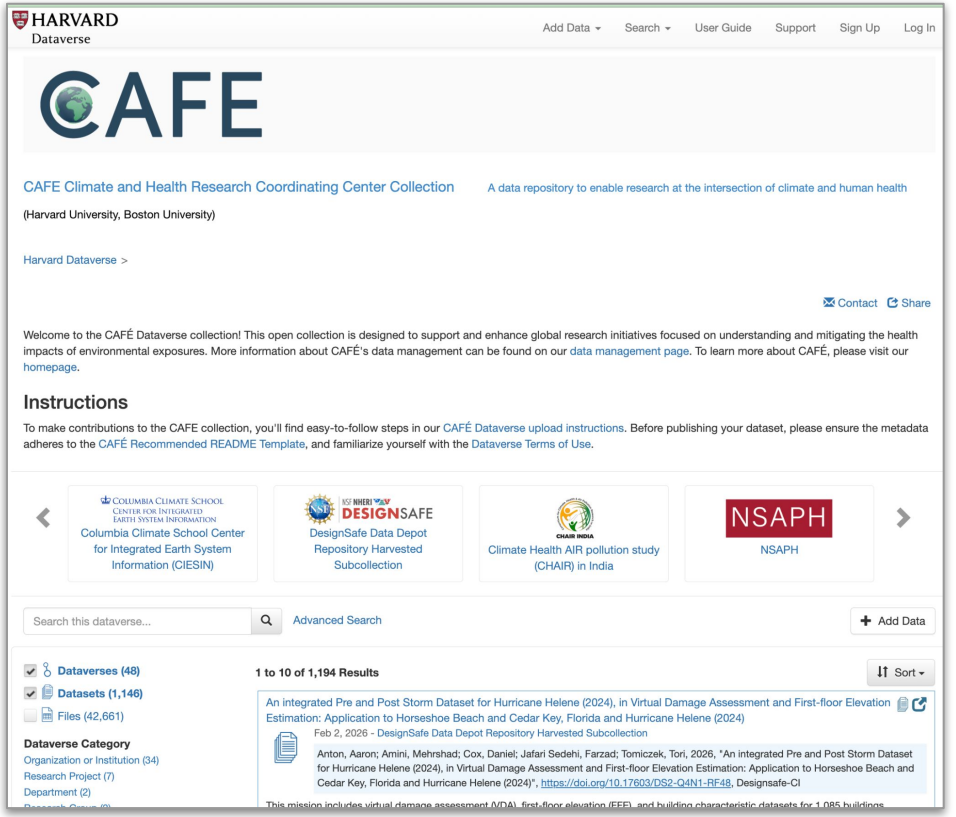
Maximum File Size

Free Storage Options	*Large Data*
Self-deposit: 2.5GB per file Contact us for larger files	Varies according to storage option

Harvard Dataverse Repository

HARVARD Dataverse

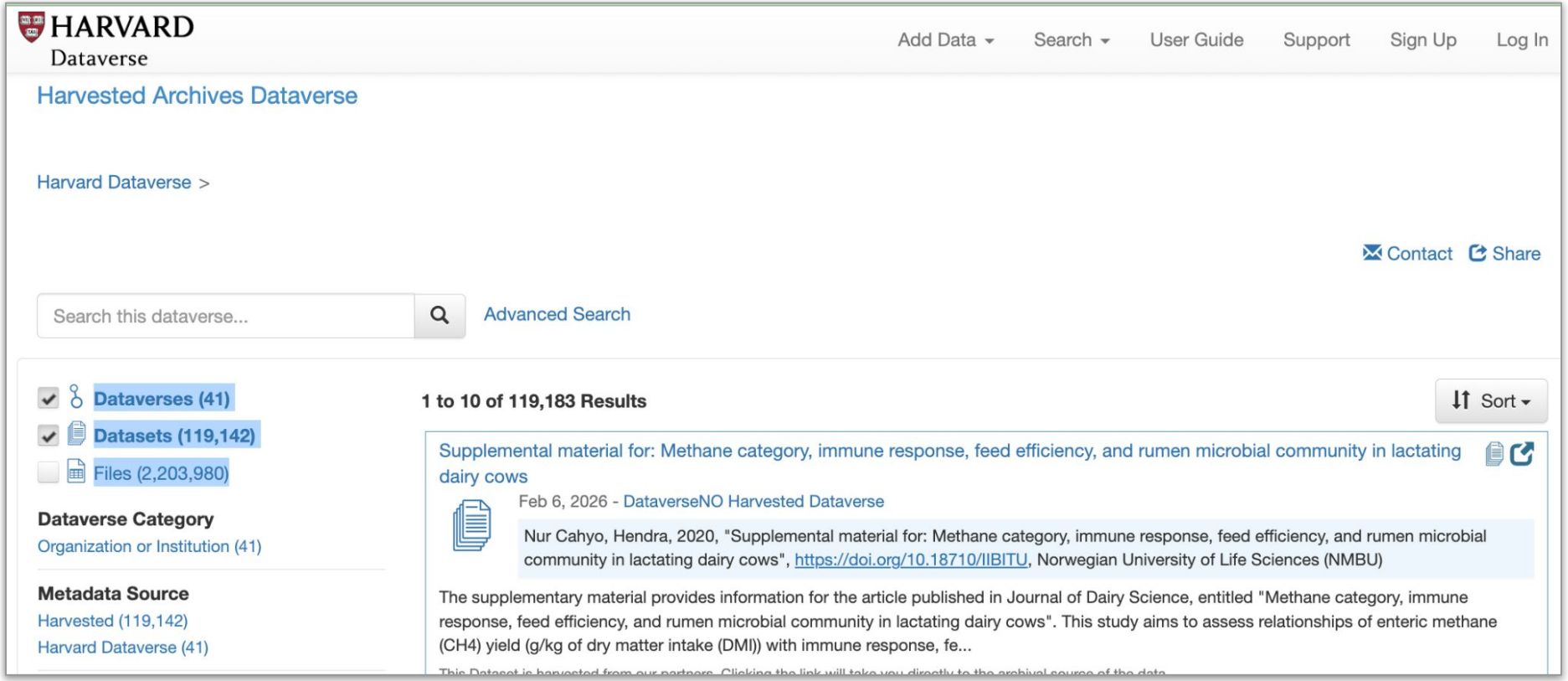




https://zenodo.org/records/18489235

Harvard Dataverse Repository

HARVARD Dataverse



Harvard Dataverse Repository

HARVARD Dataverse

Curation Services

Range of curation services, including large data

- Free and extended consultations
- Setup and maintain collections
- Upload and maintain datasets

Service Details: Link

Large Data Services

Support for large datasets (over 2.5T or millions of files)

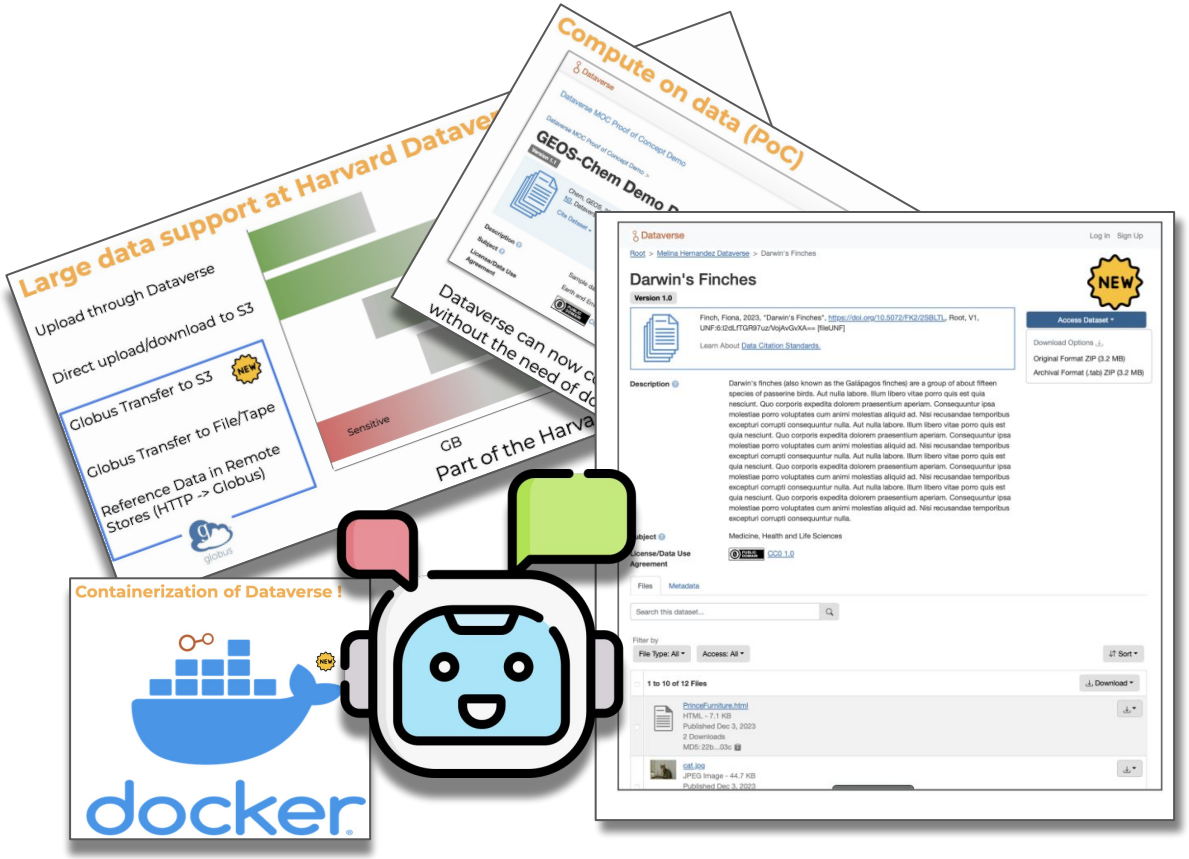
- NESE Globus Tape
- NESE Globus & S3 Disk
- AWS S3

Minimum 6 years storage

Service Details: Link

Trainings & Office Hours: Calendar
Support: support@dataverse.harvard.edu

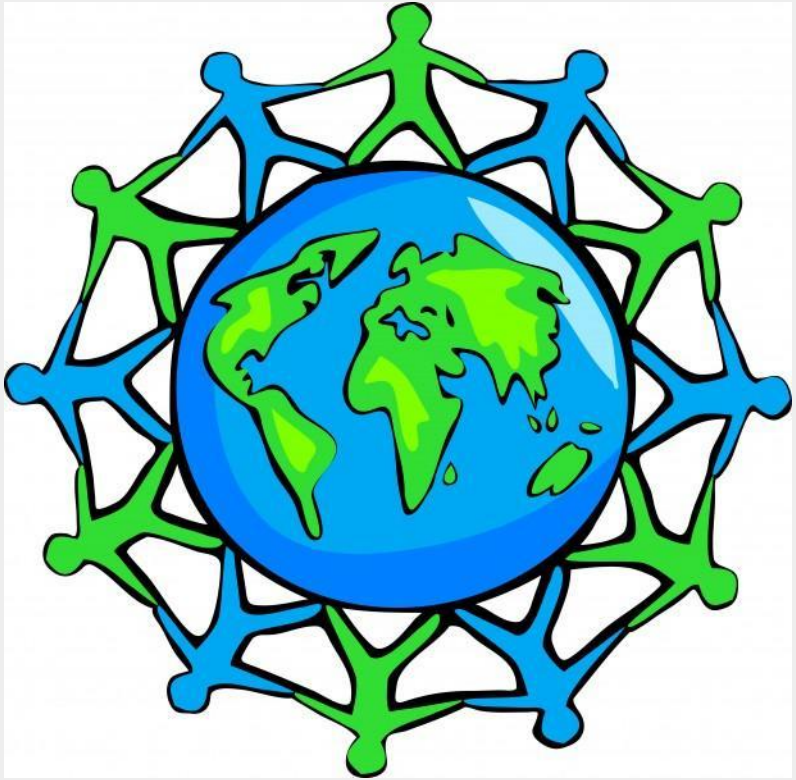
Dataverse Features: new or coming soon



- Redesigned UI
- Large data support (via Globus)
- Community-supplied dataset trust evaluations
- AI-powered user deposit tools
- Indigenous data support via TK Labels
- Locally FAIR data (restricted metadata visibility)
- Support for differential privacy tools

Global Dataverse Community Consortium

Open Network for Dataverse Users, Managers, Curators, Researchers, and Other Community Members



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About the Global Dataverse Community Consortium



- **Community Support and Coordination:** GDCC was established in 2018 to support the Dataverse community by coordinating and facilitating collaboration, community efforts, and resources worldwide.
- **Governance and Strategic Direction:** GDCC provides governance and strategic guidance to keep the Dataverse Project aligned with user needs from a diverse community.
- **Collaborative Venue:** GDCC provides a collaborative venue for institutions to leverage economies of scale in support of their Dataverse repositories around the world.

GDCC's Range of Community Services and Activities

- Providing DOIs for Dataverse installations through the GDCC DataCite consortium membership
- Supporting coordination of community development efforts
- Providing design guidance, initiating strategic redesign discussions/documents
- Development/maintenance related to community contributions/issues/projects
- Adding to the Core Developer Team (Jim Myers, Don Sizemore, Oliver Bertuch)
- Reviewing community PRs
- Helping with CI, infrastructure
- Outreach/support in using new features, managing Dataverse installations
- Security - helping alert the community of issues and providing work-arounds, patches, solutions

Let's **focus on three community activities** that GDCC facilitates:

- Community meetings
- Community calls
- Working groups

Dataverse Community Meetings



- **Annual event** since 2015, evolving from Harvard-hosted to globally distributed (Portugal, Mexico, North Carolina).
- **Main venue** for facilitating discussing past, ongoing, and future community efforts, and for fostering networking and collaboration.
- Organized by a **rotating team**: local hosts + other (more experienced) community members to support newcomers.
- Planning starts with a **community survey** to identify relevant topics (e.g., Metadata, Large Data Support, AI, Governance).
- **Open call for contributions** in diverse formats: talks, workshops, roundtables, demos, posters, WG sessions, etc.
- **Challenges:** engaging newcomers, travel limitations, and inclusivity in hybrid formats.

Working Groups: What and how

Current WGs:

- Containerization
- Documentation
- Large Data Support
- pyDataverse
- Sensitive Data



- **WGs form organically** when community members identify a shared need or goal.
- Led by **volunteer coordinators**, ensuring alignment with broader community priorities.
- **Open and inclusive membership**, welcoming diverse contributors.
- **Common characteristics:** Regular meetings, progress documentation, and updates via mailing lists and community events.
- **Challenges** include **limited capacity**, shifting leadership, and extended timelines.

Working Groups: Examples

Current WGs:

- Containerization
- Documentation
- Large Data Support
- pyDataverse
- Sensitive Data



- **Sustainability WG** – initiated by GDCC with a mandate/charter, guided by the It Takes a Village framework, produced a public report with strategic recommendations.
- **External Working Vocabulary WG** - defined a mechanism to use community vocabularies and PIDs in Dataverse, created a design document to reach consensus, developed software and a paper, now used with ORCID, ROR, keywords, etc

Community Calls



- Virtual monthly calls open to all community members.
- Platform for sharing updates, discussing issues, and showcasing new developments.
- Encourages cross-institutional collaboration and informal knowledge exchange.
- Often features guest presentations, demos, and Q&A sessions.
- Archives and upcoming call info available at dataverse.org/community-calls.

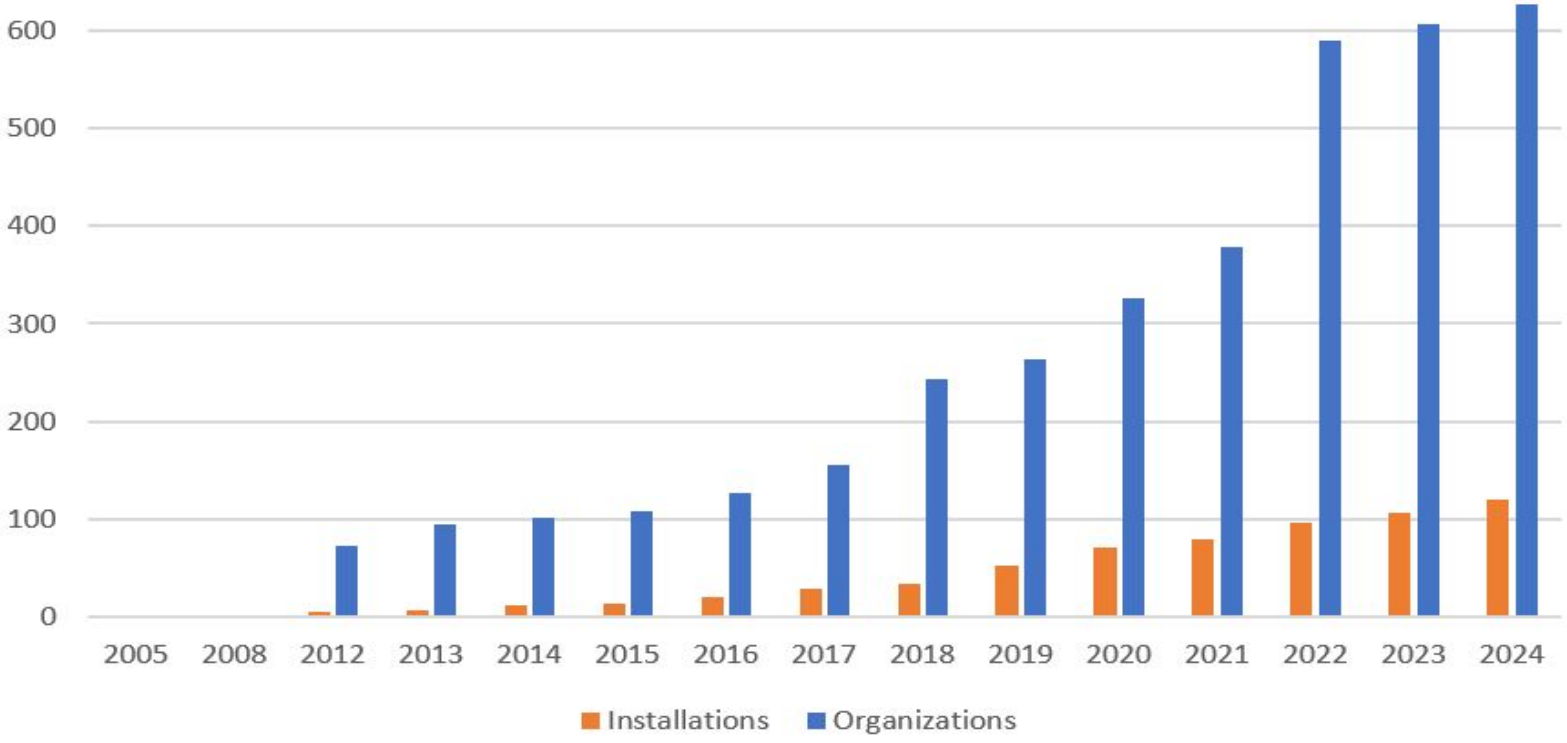
Challenges

The main challenge our community is facing is due to the **growing popularity** of Dataverse software.

The next slides illustrate some indications of the Dataverse community being **rapidly growing**.

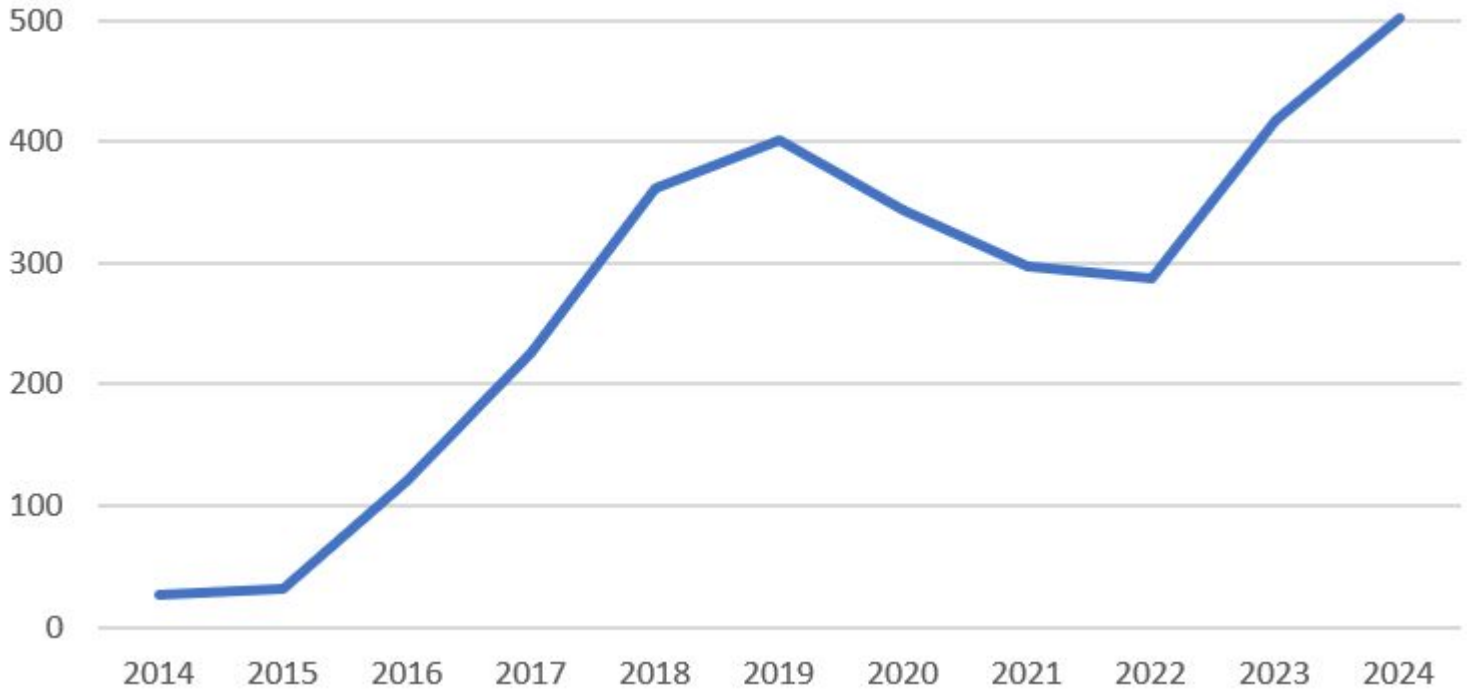


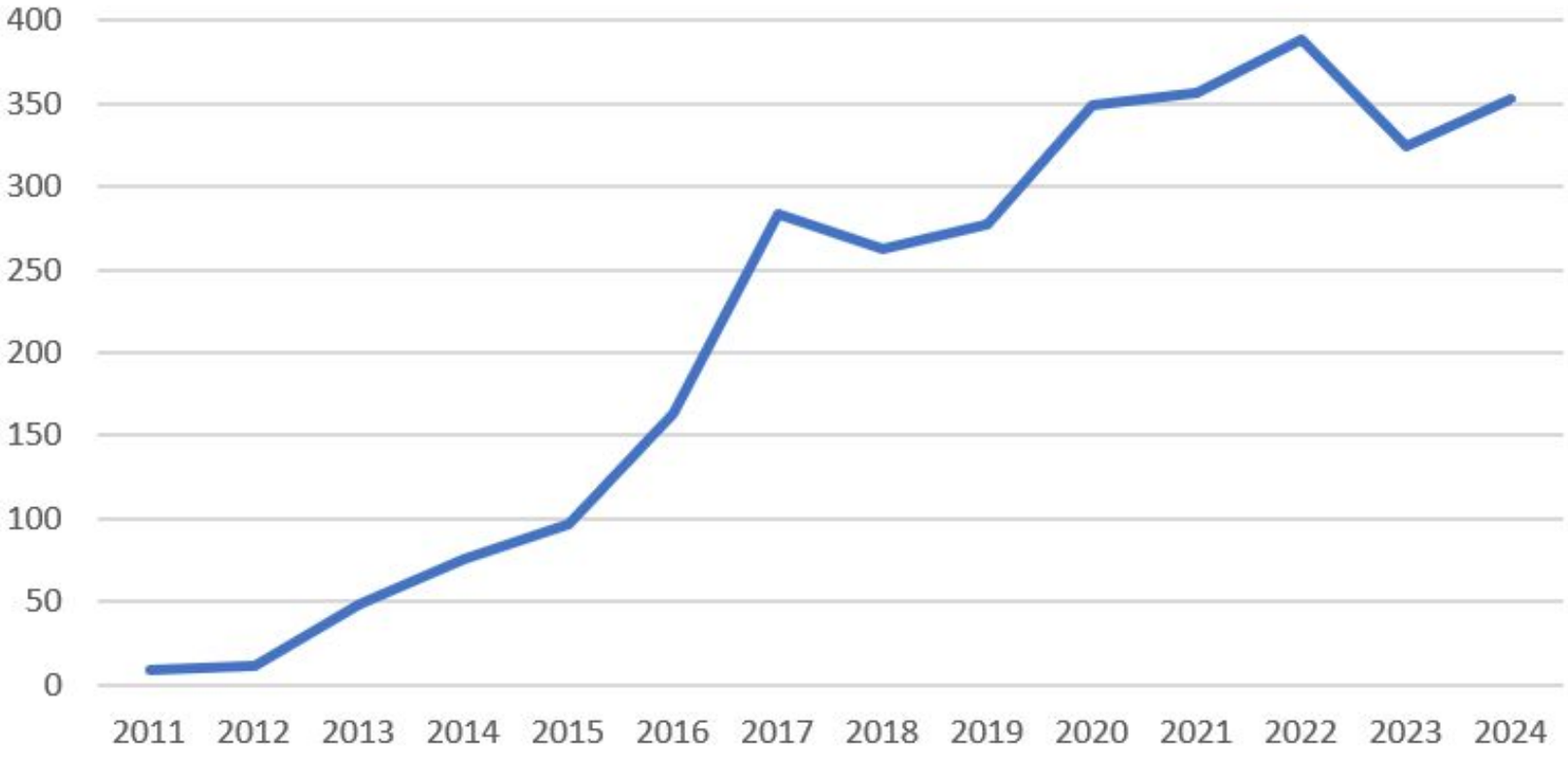
Number of Dataverse installations and served organizations (accumulated)



Source: https://iqss.github.io/dataverse-installations/

Merged GitHub pull requests IQSS/dataverse + IQSS/dataverse-frontend





Started Google Group conversations

The bright side of growth

This all testifies to

- the popularity of the Dataverse software and the ecosystem of integrated tools and services;
- people and organizations see the benefits of open-source software driven by a global community of dedicated professionals.

Challenges of growth



- The growing Dataverse community includes members with **different needs, expectations and levels of commitment** to community efforts.
- This poses some **challenges to the sustainability** of the community and the software. In particular, we need to address questions like these:
  - How to make sure the Dataverse software and its ecosystem of associated tools and services are **developed and maintained in a sustainable way to support the needs of the community**?
  - How to further **empower the GDCC** to contribute to the Dataverse community with the support and services we need?

How to address challenges of growth?

- To address these challenges, we established a **working group on sustainable growth of Dataverse** in 2023 to provide evidence-based, community-reviewed, and consensus-based recommendations on how to sustain and grow the Dataverse community.
- In October 2024, the working group delivered their **final report with 10 recommendations** on how to sustain and grow the Dataverse community.

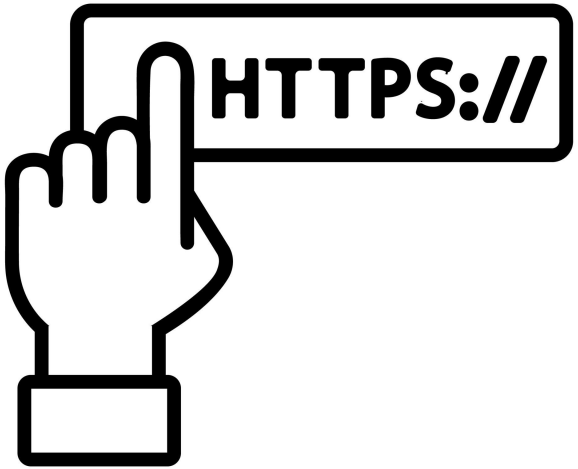
10 recommendations for sustainable growth

Source: Bloemen et al. 2024

1) Establish a new **legal organization** for GDCC
2) Establish a **formalized governance** of GDCC
3) **Clarify/demonstrate the value/benefits** of the GDCC membership and strengthen the value exchange
4) **Formalize** the **financial model** and fee contributions/membership model
5) **Diversify income streams** for GDCC
6) Establish a **functional strategy** for the Dataverse software
7) Formalize and set up a **decision-making process**
8) Make it **easier to deploy and upgrade** the application and integrations
9) Formalize and document the **communication strategy** for high-priority items
10) **Help grow, manage and sustain** a community-supported ecosystem of Dataverse-integrated tools, services, solutions, and add-ons

Resources

- The Dataverse Project
- Dataverse GitHub Repository
- The Global Dataverse Community Consortium (GDCC)
- Dataverse Community Meetings
- Dataverse Community Calls
- Dataverse Working Groups
- Harvard Dataverse Repository
- DataverseNO





Discussion